

Carbon price pass-through in Alberta's electricity market

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Carbon Pricing's Effect on Electricity Markets

- We want to understand how carbon prices impact firm offer behaviour in electricity markets
- Electricity markets, at least in certain jurisdictions, are ideal environments in which to study responses to carbon prices
- Alberta policy changes alter both carbon prices and output-based credit allocations
- Rich, hourly electricity market data allows us to observe firm-level decisions with respect to carbon price pass through
- Fabra and Reguant (2014) shows how firms respond to emissions credit prices and that lump-sum credit allocations were not distortionary in the short run
- Brown et al. (2018) have looked at the dynamics of such a market in a simplified, simulation model

Alberta's Power Market is an Ideal Laboratory

- Alberta's power market is relatively small, isolated, and features a mix of generation technologies
- Significant hourly variability in intertie capacity helps us with identification
- Alberta has minimal intertie capacity: WECC (1325 MW export and 1500 MW import path-rating) and Saskatchewan (150 MW path-rating)
- Peak internal load for 2018 was 11,697 MW, minimum internal load was 7,819 MW. Sustained low prices prevailed from 2015 through 2018.
- Supply mix (2018) was 45% coal, 21% net-to-grid from combined heat and power plants, 19% natural gas, 3% hydro, 6% wind, 1% other sources including solar and 5% imports.

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Discussion

- Alberta's energy-only power market provides an ideal environment to study carbon price pass-through
- Alberta policy changes alter both carbon prices and output-based credit allocations provide facility-specific treatment
- Preliminary results show that policy changes in 2016, 2017, and 2018 saw significant changes in offers from coal and natural gas combined cycle plants
- Framework allows us to examine changes in firm, offer control, fuel or other dimensions as well as market power implications
- Machine-learning-based estimation of synthetic offer curves due to market conditions still to come
- Glass half-full: impact of new policies and other proposed changes might provide an additional treatment.

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